ELLAN VANNIN / RESEARCH & STRATEGY

The Lough Neagh Heavy-Oil Scenario

A wealth-management assessment of a hypothetical 100-million-barrel discovery in Northern Ireland

INVESTMENT QUESTION

Could a very dense heavy-crude resource create a material new export industry for Northern Ireland, improve UK energy resilience and generate investable value?

Prepared for: Ellan Vannin Wealth Management

Research date: 12 October 2026

Status: Scenario analysis; commercial and legal verification required

PRIVATE CLIENT & INSTITUTIONAL INVESTOR BRIEFING

A disciplined distinction between resource volume, recoverable reserves and value to investors.

Investment conclusion

A potentially valuable regional asset - not yet a valuation

For this commissioned scenario, Ellan Vannin assumes that a discovery was made in summer 2026 in and around Lough Neagh and that the headline resource estimate is 100 million barrels of very heavy crude. The discovery-specific premise has been supplied for modelling; this note does not certify the discovery or its reserve classification. The economics below are conditional on verification.

> The central investment finding is simple: the discovery could matter significantly to Northern Ireland, but its financial value is determined by recoverable volume, realised price, development cost, legal permission and environmental risk - not the headline barrel count.

RESOURCE CASE	CENTRAL SALES CASE	ENERGY SCALE
100m barrels assumed resource	$5.0bn gross sales at $50/bbl if all 100m are recoverable	~0.6 EJ illustrative heat equivalent using standard crude-oil conversion

What investors should take from the scenario

1. Resource value: if 100 million barrels are commercially recoverable, gross lifetime sales at an assumed realised price of $50/bbl would be $5.0bn (about £3.85bn at an illustrative $1.30/£ exchange rate). Gross sales are not profit, tax revenue or net present value.

2. The heavy-crude nuance: “heavy” generally describes low API gravity and density/viscosity, not automatically exceptional energy content. Heavy crude may require more treatment and can sell at a discount because it produces a different mix of refined products. A lab assay and refinery-specific bids are essential. [3][4]

3. Regional impact: the strongest upside for Northern Ireland would be a locally anchored engineering, logistics and technical-services supply chain, not just extraction. Oil production is capital-intensive, so local employment and GVA depend on the share of spending captured locally.

4. Constraint: the Lough is a major water source, and a Bill to repeal Northern Ireland’s onshore petroleum licensing provisions remains in the legislative process as at 12 October 2026. The legal route and environmental acceptability are core investment gates, not footnotes. [1][2]

Assumptions used throughout

Parameter	Treatment in this note
Headline volume	100 million barrels, as a client-supplied scenario. The report distinguishes oil in place from commercially recoverable reserves.
Prices	Illustrative realised-price cases of $35, $50 and $65 per barrel. These are modelling assumptions, not a live market quote or forecast.
Currency	$1.30 = £1 solely for comparing orders of magnitude; FX is not a forecast.
Costs / recovery	Any production-cost and recovery-factor sensitivities are illustrative. No field-specific engineering study or capital estimate was available.

01 / RESOURCE & VALUATION

First question: what do the 100 million barrels represent?

In petroleum economics, “resource”, “oil in place” and “recoverable reserves” are not interchangeable. If the 100 million barrels are oil in place, only a fraction may be recoverable. If they are independently assessed recoverable reserves, the headline commercial opportunity is much larger. A resource estimate without an explicit classification cannot support a defensible asset valuation.

Gross sales sensitivity

The table below multiplies assumed recoverable barrels by an assumed average realised price. It is a gross-revenue ceiling before capital expenditure, operating costs, transport, taxes, royalties, financing and decommissioning.

Price received	100m recoverable	25m recoverable	15m recoverable
$35 / barrel	**$3.50bn** **£2.69bn**	$875m £673m	$525m £404m
$50 / barrel	**$5.00bn** **£3.85bn**	$1.25bn £962m	$750m £577m
$65 / barrel	**$6.50bn** **£5.00bn**	$1.625bn £1.25bn	$975m £750m

25m and 15m barrels are illustrative recovery outcomes if “100m barrels” instead refers to oil in place. The table does not assert a particular recovery factor. Sterling amounts use the stated illustrative exchange rate.

A realistic field value is a discounted cash-flow value

Investor value is based on future cash generated after development and operating expenditure, adjusted for the timing and risk of those flows. A simplified formulation is:

> Project NPV = present value of future operating cash flow - initial capital expenditure - abandonment and other liabilities.

A credible NPV cannot be calculated from the headline resource estimate alone. It needs a field development concept, a production curve, a crude assay, infrastructure costs, a fiscal model, an abandonment plan and a risk-appropriate discount rate. Early-stage appraisal should use ranges and probability-weighted cases rather than a single point estimate.

Energy content: meaningful, but not a substitute for commercial quality

For scale only, the US Energy Information Administration’s 2026 conversion estimate is about 5.689 million Btu per barrel of crude oil. Applying that generic factor to 100 million barrels implies roughly 569 trillion Btu, or about 0.60 exajoules of thermal energy. Actual calorific value will depend on the specific crude and the standard used. [5]

This is an energy-equivalent calculation, not a claim about useful energy delivered. Extraction, heating, dilution, upgrading, refining and combustion all affect net energy, product yield and emissions. A “heavy” label is not proof that the crude has unusually high usable energy per barrel.

Valuation read-through

If the 100m figure is a recoverable reserve, the gross sales ceiling is measured in several billion dollars under the assumptions shown. If it is an in-place estimate with a 15-25% recovery outcome, the gross sales range is smaller by roughly three-quarters to five-sixths. That classification is the first number the investment committee should seek to verify.

02 / PROJECT ECONOMICS

From a crude discovery to an investable field

The decisive commercial variable is the margin available after the crude's market discount and the cost of getting it to a buyer. Heavy crude can require more pumping, heating or dilution and may need a refinery with suitable upgrading capacity. Crude API gravity and sulphur content affect both the quality discount and processing cost. [3][4]

Illustrative field-level margin sensitivity

To expose the leverage to assumptions, the table uses three deliberately simplified cases. "Non-capital cost" means a combined placeholder for variable production, handling, transport and treatment costs; it excludes development capital, financing, tax and abandonment. These figures are not estimates for a Lough Neagh field.

Case	Realised price	Variable cost	Margin before capex/tax	Total margin on 100m recoverable
Downside	$35/bbl	$30/bbl	$5/bbl	$0.5bn
Central	$50/bbl	$25/bbl	$25/bbl	$2.5bn
Upside	$65/bbl	$20/bbl	$45/bbl	$4.5bn

The downside case demonstrates the risk: a headline multi-billion-dollar gross sales value can coexist with a very thin pre-capital margin. In the central illustration, a $25/bbl variable margin produces $2.5bn before the cost of building the field, financing, fiscal payments and closure. Those omitted costs could materially change - or eliminate - the investment return.

Production pace and indicative lifetime

Production rate	Annual barrels at plateau	Gross revenue at $50/bbl	Simple duration of 100m barrels
5,000 b/d	1.83m	$91m/year	~55 years
10,000 b/d	3.65m	$183m/year	~27 years
15,000 b/d	5.48m	$274m/year	~18 years

This arithmetic assumes steady plateau production, no downtime and no decline; it is a scale illustration, not a reservoir forecast. Real fields usually have ramp-up, decline and shutdown periods. A faster production profile brings cash forward but may require more capital and could shorten field life.

What is needed before capital can be priced?

A credible pre-FID package would include: independent resource certification; pressure and flow-test data; a laboratory crude assay; concept-select engineering; well and surface-facility design; export-route options; a risked production schedule; cost estimates with contingencies; oil-price and quality differentials; abandonment security; and legal/environmental consents.

No refinery should be assumed

Northern Ireland does not currently have an operating crude-oil refinery. A base-case business plan should therefore price the crude for export to an identified compatible refinery, subject to transport, storage and buyer acceptance. A local upgrader or refinery is an independent capital project, and should not be treated as an automatic multiplier of the resource's value.

03 / ECONOMIC TRANSMISSION

What it could mean for Northern Ireland and Britain

The likely regional impact is more meaningful than the national macro impact. At UK scale, a field with a 10,000-barrel-per-day plateau would be a useful marginal source of supply, but too small to determine global oil prices or remove the need for imports. For Northern Ireland, the same field could become a material export business and a platform for specialist industrial capability.

Northern Ireland: where the value could accrue

Direct effects would include exploration and production work. Wider value could accrue through civil engineering, equipment maintenance, metrology, industrial software, environmental monitoring, logistics, storage and technical education. The lasting benefit would depend on how much expenditure is won by local firms and how much specialist equipment, expertise and finance must be imported.

The project should be judged on regional value added, not the gross value of oil produced. A $183m annual sales stream in the 10,000 b/d illustrative case is not £140m of Northern Irish GDP: sales include imported inputs, services, operating costs and other payments, while profits may accrue to owners outside the region.

UK-wide channels

Supply diversification. Local production could add another source of crude to a market still connected to internationally traded supplies. The strategic gain is incremental resilience rather than energy independence. [6]

Trade balance. Exports or displacement of equivalent imports could improve external trade, but the net gain depends on where the crude is sold, what imports are displaced, and the imported capital and operating inputs needed to produce it.

Refining and product prices. Domestic production does not automatically lower petrol, diesel or jet-fuel prices. The crude may be sold at export parity and refined elsewhere; any consumer benefit depends on the market and refinery economics.

Fiscal receipts. Tax, royalties or other public receipts would depend on the legal regime and project profitability. They should not be estimated by applying a headline percentage to gross sales. The actual fiscal design and liability allocation require legal and tax analysis.

The strongest regional-development strategy

If the project passed all gates, Northern Ireland could seek a competitive local services cluster around drilling support, equipment reliability, data and automation, environmental services, materials handling and specialist training. These skills could remain useful beyond the field's producing life. Procurement should be transparent and competitive: mandating local spend without building capable suppliers risks raising costs without building productivity.

A resource-transition fund could also be considered if public revenues materialised. It should receive only actual cash receipts and have strict governance, transparent investment rules and a mandate to convert temporary resource income into long-lived public assets, skills or remediation. It cannot sensibly be capitalised on the field's gross sales estimate.

A caution from previous policy analysis

A 2021 report prepared for Northern Ireland's Department for the Economy modelled the regional effects of broader onshore petroleum-development scenarios and found that outcomes depended heavily on the scale of activity and the share of expenditure retained locally. Its scenarios were not an assessment of this hypothetical 100m-barrel field, so they should inform questions, not be mechanically scaled to produce a forecast. [7]

04 / RISK & FEASIBILITY

The risks that could determine the whole outcome

The potential economic upside is conditional on a narrow set of issues being resolved. At Lough Neagh, the legal and environmental gates are unusually important because the lake is a major drinking-water source: Northern Ireland Water says treatment works using its raw water supply around 40% of the population. [2]

Risk	Why it matters	Evidence / diligence required	Investment response
Reservoir / recoverability	Oil in place may be much larger than recoverable volume; heavy viscosity may constrain flow.	Independent competent-person report, pressure tests, permeability, recovery method and decline curve.	Do not capitalise headline barrels before reserve classification.
Crude quality / market	Density, viscosity, sulphur, metals and contaminants affect price, treatment and buyer set.	Full assay, sample testing, refinery compatibility and binding/indicative offtake bids.	Use grade-specific realised-price assumptions, not Brent alone.
Capital / cost overrun	Drilling, heating, dilution, export systems and water management may be costly.	Concept-select engineering, estimate maturity, contingency, vendor quotes and sensitivity runs.	Stage capital; use independent technical assurance.
Law and licensing	The current licensing framework is under legislative change.	Northern Ireland counsel's written opinion on the law in force, transitional position and any lawful route to activity.	Treat legal authorisation as a hard gate, not a probability-weighted afterthought.
Water / environment	A spill or groundwater impact could impose public, ecological, remediation and reputational costs.	Baseline hydrogeology, independent impact assessment, credible containment, emergency and closure plans.	No-go if risk cannot be reduced to acceptable and enforceable levels.
Fiscal / closure liability	Taxes, royalties, decommissioning security and remediation could absorb much of project value.	Fiscal model, liability allocation, bonding/security and abandonment cost estimate.	Value the full life-cycle obligation, not just production years.

Legal position as at 12 October 2026

The Northern Ireland Assembly's official bill page records the Petroleum Exploration and Licensing (Repeal) Bill as introduced on 13 April 2026, with Committee Stage completed on 7 October. Consideration Stage and later steps were listed as still to be scheduled, and Royal Assent had not been recorded on the page reviewed. The bill would repeal provisions relating to onshore petroleum exploration and licensing. This note does not provide a legal opinion; the project's exact position would need to be checked with Northern Ireland counsel against legislation and policy in force on the relevant date. [1]

Environmental conditions are not a monetisable externality to ignore

The area's water-supply role means independent baseline data, well-integrity assurance, monitoring, containment capacity and long-term remediation security would be central to the investment decision. An environmental incident could impose costs far beyond the operator's direct footprint. A project whose economics only work by underpricing this risk is not a robust investment.

05 / INVESTMENT COMMITTEE FRAMEWORK

A staged approach: verify before valuing

Ellan Vannin should not make an investability judgement from the headline alone. A disciplined process would progress through defined gates, with each gate requiring evidence before further capital or reputational commitment.

Gate	Evidence required	Decision test
1. Discovery verification	Public technical announcement; independent resource report; precise location; well and test data; classification of 100m barrels.	Is the claimed quantity evidenced, and is it oil in place, contingent resource or recoverable reserve?
2. Crude and recovery	Assay for API gravity, viscosity, sulphur and metals; flow testing; production method; recovery factor.	Can the field produce enough saleable crude at a sustainable rate?
3. Commercial route	Target-refinery list, export logistics, sample acceptance, marketing differential and costed transport.	Is there an identified buyer at a defensible netback?
4. Legal and environmental	Written legal opinion; consent pathway; independent water/hydrogeology assessment; risk controls and secured liabilities.	Is there a lawful and environmentally defensible route to production?
5. Full investment case	Risked DCF, capital estimate, tax and fiscal model, scenario analysis, abandonment security and funding terms.	Does risk-adjusted NPV and downside protection justify capital at risk?

Ellan Vannin's base-case view

The discovery, if verified as described, could represent a multi-billion-dollar gross sales opportunity across its producing life. But the gross-revenue number is not an asset valuation. Under the illustrative central operating margin, 100 million recoverable barrels create $2.5bn of total variable margin before development capital, financing, tax and closure costs; a downside margin assumption reduces that figure to $0.5bn. If only 15-25 million barrels are ultimately recoverable, economics contract sharply.

For Northern Ireland, value creation would be strongest if project spending builds a competitive local industrial supply chain and if any public receipts are preserved for long-lived benefits. For the United Kingdom, the main gains would be a modest increase in domestic supply, trade diversification and technical activity - not energy independence or a material reduction in global oil prices.

Final judgement

> Treat the 100-million-barrel case as a potential strategic resource, not a proven financial asset. The investment case becomes compelling only if independent evidence establishes recoverable reserves, a marketable crude grade, robust field economics and a lawful, environmentally acceptable development route. In and around Lough Neagh, failure on the legal or water-protection test can overwhelm an otherwise attractive gross resource value.

Scope & important notice

This report is a scenario-based economic note prepared in the name of Ellan Vannin Wealth Management for discussion purposes. It is not a verified reserves report, engineering feasibility study, valuation, legal opinion, tax opinion, offer, recommendation to transact, or investment advice. Figures not specifically attributed to external sources are illustrative modelling assumptions. No discovery-specific claims have been independently authenticated in this note. Any real decision requires qualified petroleum engineering, environmental, legal, fiscal and financial due diligence.

APPENDIX / SOURCES & METHODOLOGY

Selected source register

External references consulted for the background facts and definitions used in this note. Discovery volumes, prices, recovery factors, costs and production cases are treated as scenario assumptions unless otherwise stated. Web pages checked on 12 October 2026.

[1] Northern Ireland Assembly. Petroleum Exploration and Licensing (Repeal) Bill - official bill progress and legislative stages. Accessed 12 October 2026.
https://www.niassembly.gov.uk/assembly-business/legislation/2022-2027-mandate/primary-legislation-bills-22-27-mandate/petroleum-exploration-and-licensing-repeal-bill/

[2] Northern Ireland Water. Lough Neagh Blue-Green Algae - raw-water abstractions and treatment works serving around 40% of the population of Northern Ireland. Accessed 12 October 2026.
https://www.niwater.com/about-your-water/bga

[3] U.S. Energy Information Administration. Definitions, Sources and Explanatory Notes: Qualities of Crude Oil Input - API gravity and heavy-crude classification. Accessed 12 October 2026.
https://www.eia.gov/dnav/pet/TblDefs/pet_pnp_crq_tbldef2.asp

[4] Natural Resources Canada. The Refining Sector in Canada - relationship between crude density, sulphur, product yields, refinery configuration and heavy-crude processing cost. Accessed 12 October 2026.
https://natural-resources.canada.ca/energy-sources/fossil-fuels/refining-sector-canada

[5] U.S. Energy Information Administration. Energy conversion calculators - crude oil conversion factor (2026 estimate: 1 barrel = 5.689 million Btu). Accessed 12 October 2026.
https://www.eia.gov/energyexplained/units-and-calculators/energy-conversion-calculators.php

[6] Department for Energy Security and Net Zero. Digest of UK Energy Statistics (DUKES): Petroleum - official source for UK oil supply, demand, imports and exports; 2026 edition updated 30 July 2026.
https://www.gov.uk/government/statistics/petroleum-chapter-3-digest-of-united-kingdom-energy-statistics-dukes

[7] Hatch Regeneris. Economic Assessment of Onshore Petroleum Licensing in Northern Ireland, report commissioned by the Department for the Economy, 2021. Used only as context on the importance of scale and local supply-chain retention; not as an estimate for this hypothetical field.
https://www.economy-ni.gov.uk/sites/default/files/consultations/economy/Hatch-Regeneris-OPL-report.PDF

Methodology notes

Gross sales = recoverable barrels × realised price. Sterling equivalents assume $1.30 per £1 and should not be interpreted as a current FX quote. Energy equivalent applies an EIA generic crude-oil heat-content conversion to the headline volume; actual assay data may differ. Illustrative production durations divide assumed recoverable barrels by a constant plateau rate and ignore decline, downtime and ramp-up. Field-level margins subtract illustrative variable operating/transport/treatment cost from assumed realised price and remain before capital expenditure, taxes, financing and abandonment. No probabilistic weighting is performed because field-specific data are not available.

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